Clinton Enwerem

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Dear Technical Recruiter,

Application: Research Scientist, Robotics

I work on dexterous manipulation, robot learning, motion planning, state estimation, and feedback control. I am an Electrical and Computer Engineering Ph.D. candidate at the University of Maryland, advised by Dr. John Baras and Dr. Calin Belta and graduating in Fall 2026. I formulate technical questions, implement the resulting methods in Python and C++, and evaluate them in simulation and on physical arms with dexterous hands.

In ParcelStow, I built an Isaac Lab Unitree G1 and RealHand L6 grasp–reorient–insert task, collected 297 successful expert demonstrations, and trained DAgger, Diffusion Policy, and ACT. The expert and one ACT policy both reached 100% nominal success. At twice nominal speed, expert success was 84% and ACT success was 53%, a 31-percentage-point difference within the demonstrated speed range. Stage outcomes placed 35 of ACT's 47 failures during insertion.

My manipulation research also addresses uncertain contact and feedback execution. FIRMGrasp uses a CVaR wrench-space margin to rank grasps under friction uncertainty. Grasp Distance Fields use a CBF-CLF quadratic program during execution; simulated experiments lifted 46 of 50 objects, retained a median 94% of the certified grasp margin, and solved the program in 0.09 ms. In EquiDexFlow, I developed a flow-matching grasp generator and retargeted its predicted contacts to a physical LEAP Hand. I fine-tuned OpenVLA-7B with QLoRA for recovery supervision in simulated bimanual packing. Each of three training seeds completed 136/200 tasks versus 129/200 without supervision, recovering seven failed executions without disrupting nominal successes. I developed a visuo-tactile bimanual assembly pipeline with simulation and hardware execution interfaces, combining depth-based pose estimation (RANSAC/DBSCAN, ICP), collision-constrained trajectory optimization (CasADi/IPOPT), and gravity-compensated position control with fingertip-contact monitoring. I evaluated task execution in MuJoCo.

I developed dexreach to connect GPU-accelerated dexterous grasp synthesis and cuRobo motion planning with Isaac Lab sequence validation, evaluating object motion, task-stage completion, and contact-loss stops.

My resume, research statement, and curriculum vitae are attached. ParcelStow code and data and models are available online. You can reach me at **enwerem@umd.edu** or **301-416-9430**.

Thank you for considering my application. I am available to start in December 2026. I look forward to discussing how my experience in robot learning, simulation, and feedback control could support your team's work.

Sincerely,

Clinton Enwerem
October 27, 2026